2.0 Problem-Solving Concepts MA1905

2.1 Define problem-solving

In the simplest terms, problem solving means solving a problem. However, what is the problem, and how do we go about solving it?

A problem is a situation that requires unknown information to be found or a question to be answered. In mathematics, we often encounter them as word problems, in which a problem is presented as text rather than in mathematical notation.

Therefore, problem-solving is the process of identifying a problem, analyzing it, developing possible solutions, applying an appropriate solution, and checking the result.

When solving word problems, follow these general guidelines:

- Read the problem carefully and figure out what you are asked to find.
- Identify the unknown values and assign each of them a variable.
- Translate the words into an algebraic equation.
- Solve the equation using the rules of algebra.
- Answer the original problem.
- Check your result to ensure it makes sense.

2.2 Discuss types of problems

Well-Defined Problems

A well-defined problem has a clear starting point, a clear goal, enough information to work with, and rules or procedures that can be followed.

For example, a 1 200 MB file is transferred at 40 MB per second. How long will the transfer take?

Time = File Size / Transfer Rate

Time = 1 200/40 = 30 seconds

Ill-Defined Problems

An ill-defined problem does not have one obvious solution. Information may be missing, and different solutions may be reasonable.

For example, what is the best computer for a college student?

The answer depends on factors such as budget, required software, portability, storage, battery life, and operating system.

2.3 Differentiate between problem-solving with computers and general problem solving

General Problem-Solving

General problem-solving may use experience, judgment, observation, communication, research, calculations, and trial and error.

Problem-Solving With Computers

Computers are especially useful when a problem involves large amounts of data, repeated calculations, searching or sorting, or clearly defined rules that must be followed many times.

For example, a company must calculate overtime pay for 2 000 employees. A computer can apply the same rules quickly and consistently to all employees.

A computer does not automatically understand what the user intends. Appropriate data and clear instructions must be given.

Examples

Would a computer be particularly useful?

1. Calculating the average mark for 10 000 students.
2. Choosing which photograph looks best for a personal profile.
3. Deciding which applicant would be the best fit for a small project team.
4. Searching a database containing one million customer records.

2.6 Develop expressions and equations

To solve word problems, we must first know how to translate English phrases into algebraic expressions. Certain words provide us with cues as to which mathematical operations are to be used. These words appear frequently in phrases and we shall call them cue words.

Phrases Describing **Addition**	Phrases Describing **Subtraction**
Two **more than** a number The **sum of** a number and two A number **increased by** two Two is **added to** a number Two **greater than** a number A number **plus** two The **total of** two and a number	A number **decreased by** 5 Five **less than** a number. Five **smaller than** a number. Five **fewer than** a number. A number **diminished by** five. A number **minus** five. The **difference between** a number and five.
Phrases Describing **Multiplication**	Phrases Describing **Division**
Double a number. **Twice** a number. The **product** of ___and a number. **Triple** a number. Three **times** a number.	A number **divided by** 6. One-**sixth** of a number. The **quotient** of a number and six.

When translating English phrases into algebraic expressions, we translate words which represent:

- unknown quantities into variables
- numbers and operations into their algebraic symbols.

Examples

Write an algebraic expression.

1. Four more than a number.
2. Triple a number.
3. Eight fewer than a number.
4. One-half more than a number.
5. The sum of four times a number and seven.
6. Three less than the quotient of a number and four.

Sometimes, more than one operation can be performed on an unknown at the same time. In this case, we must be extra careful to ensure that the order of operations is the same as was intended. This may require the use of parentheses to enforce that order.

7. A number is increased by 8 and then it is tripled.
8. Four times the sum of a number and ten.
9. Eight more than triple a number.
10. A number is increased by eight and then it is tripled.
11. One-third of the sum of a number and four.
12. Eight divided by the difference between a number and six.

Solving Word Problems Using Algebra

We will start with the simplest kind – straight number problems. In this type, write an expression as we did on the previous page, but now we add in cue words for "=" such as "is" or "is the same as". This will allow us to write an equation that we can solve.

Examples

Four more than a number is 12. What is the number?

Eight fewer than a number is 32. What is the number?

Two-fifths of a number of 8. What is the number?

Six times a number is 15. What is the number?

Three more than 8 times a number is 99. What is the number?

Continuation of Solving Word Problems Using Algebra

Three times a certain number increased by 5 is equal to 9. What is the number?

Two fewer than twice a number is 4 more than that number. What is the number?

Three more than twice a number is that number. What is the number?

If you add 10 to a certain number, the result is the same as if you add 4 to three times the number. What is the number?

Consider two numbers. The second number is five fewer than triple the first number. The sum of the two numbers is fifteen. Find each number.

Two consecutive numbers have a sum of 91. What are the numbers?

A student pays $4 900 for tuition and housing for a semester. If the cost of housing was $500 less than twice the cost of tuition, how much did the student pay for each expense?

Bill charges a base rate of $165 plus $55 an hour to do taxes. If a client's total bill was $440, how many hours did Bill spend doing the client's taxes?

The cost of renting a car is $89 a day plus $0.25 per kilometer. How far can Marcia drive in one day if she only has $110?

The total cost (after tax) of a USB drive and a printer cartridge is $46.17. If the USB is $3.91 more than the cartridge before HST of 15%, determine the list price of each item.

Solve Word Problems Involving Inequalities

Before solving inequalities, it is important to recognize some key words.

English Phrases Describing Inequality	Algebraic Expression
Greater than "…is more than…" "…is larger than…"	
Less than "…is fewer than…" "…is smaller than…"	
Greater than or equal to "…is not less than…" "…is at least…" "…cannot be less than…" "…a minimum.."	
Less than or equal to "…is not greater than…" "…is at most…" "…is not more than…" "…does not exceed…" "…a maximum…"	

Translate each English expression into an inequality. Use x as the variable.

a) The speed limit on the Outer Ring Road is a maximum of 100 km/h.

b) The minimum wage in NL is $15.60 per hour.

c) A person's systolic blood pressure is classified as high if it is above 130.

d) A grade below 70% is considered a failing grade in the Veterinary Technician Program.

Problems containing words or phrases such as at least, at most, between, no more than and no less than often require an inequality rather than an equation.

Examples

Solve using inequalities.

You need to buy lamps and curtains for your apartment. You have a maximum of $340 to spend. A set of curtains costs $48, and each lamp costs $35. If you buy 4 sets of curtains, what is the maximum number of lamps you can afford? Assume the prices include taxes.

A taxi charges a flat rate of $2.75 plus $0.50 per kilometre travelled. Determine how far a customer can travel if they want to spend less than $10.

The cost, C, of producing x coffee cups is given by $C = 105x + 900$. The revenue, R, generated from selling x coffee cups is given by $R = 115x$. Determine the values of x for which the company makes a profit.

The length of a rectangle is 3 metres more than twice its width. If the perimeter of the rectangle is at least 30 metres, write and solve an inequality to determine the possible values of the width.

A tank initially contains 400 L of water. Water drains from the tank at a rate of 1.25 L/min.

a) Write an expression for the amount of water remaining in the tank after t minutes.

b) For how long will the tank contain more than 100 L of water? Set up and solve an inequality.

General Computer-Style Notation

Up to this point, we have been writing equations and expressions using standard mathematical notation, including symbols such as +, −, ×, ÷, and √. When mathematical expressions are entered into a computer program, different symbols and notation are often needed.

For example, an asterisk (*) is commonly used for multiplication and a slash (/) is commonly used for division. Mathematical expressions must also be written on a single line, so parentheses are often needed to show which operations should be performed together.

The following notation will be used throughout this course:

Mathematical Operation	Written Notation	Computer Notation
Addition	$a + b$	a + b
Subtraction	$a - b$	a - b
Multiplication	$a \times b$	a * b
Division	$a \div b$	a / b
Exponentiation	a^b	a^b
Square root	$\sqrt{a}$	sqrt(a)
Less than	$a < b$	a < b
Less than or equal to	$a \leq b$	a <= b
Greater than	$a > b$	a > b
Greater than or equal to	$a \geq b$	a >= b

Note: The exact notation used may vary depending on the programming language. The notation shown above is the notation we will use in this course.

Let's look at some examples of writing expressions in computer notation.

Written Expression	Computer Notation	Notes
$5 \div 3$	5 / 3	The “slash” is commonly used to indicate division or fractions
7×9	7 * 9	The asterisk is commonly used to indicate multiplication.
4x	4 * x	There must always be an asterisk where multiplication is performed.
$\frac{1}{2}x + 5$	(1 / 2) * x + 5	The parentheses around 1 / 2 make it clear that only the 2 is in the denominator.
$\frac{1}{2x} + 5$	1 / (2 * x) + 5	The parentheses make it clear that both 2 and x are in the denominator.
$\frac{1}{2x + 5}$	1 / (2 * x + 5)	The parentheses make it clear that the 2 * x + 5 is in the denominator.
$\frac{x + 2}{x^2 + 5x + 6}$	(x + 2) / (x^2 + 5 * x + 6)	Parentheses are used to clearly indicate the numerator and denominator.
$\frac{\frac{x + 3}{5}}{\frac{x - 4}{2}}$	((x + 3) / 5) / ((x - 4) / 2)	Use parentheses to clearly indicate the numerator and denominator of each fraction. Using extra spaces is helpful, too.
x^2	x^2	The caret, ^, usually Shift + 6 on the keyboard, is used in this course to indicate exponents.
$x^{\frac{2}{3}}$	x^(2 / 3)	Parentheses are needed to show that 2 / 3 is the entire exponent. Without them, x^2 / 3 means x^2 divided by 3.
$x^2y^3z^4$	x^2 * y^3 * z^4	There must always be an asterisk where multiplication is performed.
$b^2 - 4ac$	b^2 – 4 * a * c	There must always be an asterisk where multiplication is performed.
$\sqrt{2y}$	sqrt(2*y)	The abbreviation sqrt stands for square root, and the parentheses make it clear that both the 2 and the y are inside the radical.

Examples

Convert the expressions from written notation to computer notation.

$a \div b$

$5a^3$

$\sqrt{5x+1}$

$3x^2 - 4x + 7$

$$\frac{x^2 + x + 3}{x^2 + x}$$

$$\frac{x}{4 + \sqrt{x}}$$

$(-32y^{10})^{\frac{1}{5}}$

Convert the expressions from computer notation to written notation.

2 * x * y – z^3

(2 * x^2) / 3

4 * x + x * sqrt(y)

(x + 3)/ x – (x - 1)/(x + 3)

2.4 Distinguish between constants and variables

Variables

In a computer program, data is stored in variables. You are already familiar with the term variable from algebra, where a variable can represent an unknown value. For example, in the Pythagorean Theorem, $a^2 + b^2 = c^2$, the letters a, b, and c are variables. In this formula, a and b represent the lengths of the two shorter sides of a right triangle, while c represents the length of the hypotenuse.

However, variables in programming are somewhat different from variables in mathematics. In programming, a variable is a named storage location in a computer's memory where data can be stored. A variable can hold data entered by the user, the result of a calculation, or another piece of information needed by the program.

The data stored in a variable is called its value. This value can change while a program is running. In this way, variables allow a computer program to "remember" and work with data.

For example, in a computer program, you may see a line of code such as this one:

```
area = pi * r * r
```

This may look like a mathematical equation, but it is not. It is an assignment statement - an instruction that tells the computer to calculate a value and store the result.

The general form of an assignment statement is:

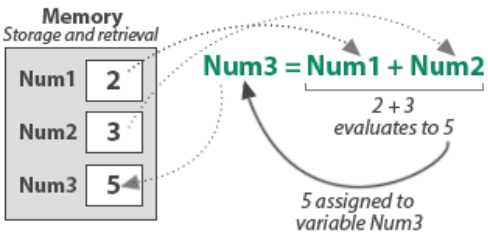
```
variable = expression
```

There is a single variable on the left side of the equals sign and an expression on the right. When the computer processes the statement, it:

- Evaluates the expression on the right.
- Stores the resulting value in the variable on the left.

Therefore, in programming, the equals sign in an assignment statement does not mean "is equal to" in the same way it does in mathematics. Instead, it means "assign the value on the right to the variable on the left."

This process is illustrated below using the assignment statement Num3 = Num1 + Num2.



Now consider the following assignment statement:

x = x + 1

If we interpret this as an algebraic equation, it makes no sense - no number can be equal to itself plus one. However, this is an assignment statement, not an equation.

The computer processes the statement from right to left:

- Retrieve the current value of x.
- Add 1 to that value.
- Store the result back in x.

For example, if x contains the value 0 before the assignment statement is executed, x will contain the value 1 afterward.

Constants

A constant is like a variable because it is used to represent a value in a computer program. The important difference is that the value of a constant does not change while the program is running.

For example, consider the calculation:

area = pi * r * r

In this statement, r may be a variable because its value can change depending on the radius of the circle. The value of pi, however, does not change. Therefore, pi can be defined as a constant.

For example: pi = 3.14159

Once pi has been defined as a constant, the program can use its value whenever it is needed without changing it.

Constants are useful for values that should remain the same throughout a program. Examples might include a tax rate, the number of days in a week, or a fixed conversion factor.

The main difference between a variable and a constant is:

	Variable	Constant
Value	Can change while the program is running	Does not change while the program is running
Example	radius = 5	PI = 3.14159
Purpose	Stores data that may change	Stores a fixed value

For example, suppose a program calculates the area of different circles. The value of the radius may change each time a new circle is entered, but the value of pi remains the same.

Examples

Determine whether the statement is a proper assignment statement. If it is not, explain why.

	Proper	Reason
X = 2		
A + 3 * 2		
9 = Y		
D = C – 2		
4 * A = 7 * B		
C = C + 1		

2.5 Describe data types

Variables can be assigned a specific data type by the programmer. A variable's data type determines the type of value that can be stored in the variable and the operations that can be performed on it.

Some of the most common data types used in computer programs are:

- Integer
    - Used for whole number.
    - It can be positive, negative, or zero.
    - Tthere is no fractional or decimal part.
    - Examples: 5, 0, -12
- Floating-Point
    - Used for numerical values that can contain a decimal point or fractional part
    - Example: 2.345, -4.5, 10.0
- String
    - Used for text.
    - String values are usually surrounded by quotation marks.
    - Examples: "Hello", “John”, “10”
- Boolean
    - Can be one of two values: True or False.
    - We will learn more about Boolean values later in this course.

For example, if you want to create a variable to store a person's name, you will use a String data type. A person’s age might be stored as an Integer, while a value such as price might be stored using a Floating-Point data type.

Examples

Determine the correct data type.

a) 0.0065

b) 1234.

c) "false"

d) false

e) 1.99

f) 25

g) “25”

h) -47

i) “A102”

Write an assignment statement for each of the following.

a) Calculate the average of 3, 9, and 12.

b) The cost of attending a movie is $12 for an adult and $7 for a child. Calculate the total cost for a group of x adults and y children to attend the movie.

c) A program stores a person's name, "Alex", in a variable called name. Write the computer assignment statement.

MA1905 – Unit 2 Practice

1. Translate each phrase into an algebraic expression. Use x as the variable.

a) Nine less than twice a number.

b) Three times the difference between a number and 7.

c) The quotient of a number and 5, increased by 4.

d) Two-thirds of the sum of a number and 12.

2. Convert each expression to computer notation.

a) $8x^3 - 5y$

b) $\frac{2x-7}{x+4}$

c) $\sqrt{5x-1}$

d) $x^{\frac{5}{2}} + 3xy$

3. Convert each expression from computer notation to standard written mathematical notation.

a) 6 * x^3 + 2 * x

b) (4 * y - 1) / (y + 6)

c) sqrt(x + 9) - 2 * y^2

4. Write a computer assignment statement for each situation.

a) Store the product of length and width in a variable called are

b) Calculate the total cost of n tickets at $18.50 each and store the result in total.

c) Calculate the average of test1, test2, test3, and test4 and store the result in average.

d) Store the text "Pending" in a variable called status.

5. Translate each statement into an inequality. Use x as the variable.

a) A package may weigh at most 23 kg.

b) A mark below 60% requires reassessment.

c) A user must be at least 18 years old.

d) A temperature greater than 85°C causes an alert.

6. Set up and solve an appropriate equation for each of the following. Clearly indicate what the variable represents. Show all steps and state your final answer.

a) A company ordered 42 computer monitors and keyboards altogether. The company ordered 8 more monitors than keyboards. How many monitors and how many keyboards were ordered?

b) A student purchased a computer accessory. After 15% sales tax was added, the total cost was $172.50. What was the price of the accessory before tax?

c) Three consecutive whole numbers have a sum of 204. Find the three numbers.

d) A rectangular room is 5 m longer than it is wide. Its perimeter is 50 m. Find its dimensions.

7. Set up and solve an appropriate inequality for each of the following. Clearly indicate what the variable represents. Show all steps and state your final answer.

a) A student has already saved $275 toward a new computer. The student saves $45 each week and wants to have at least $680. How many weeks must the student continue saving?

b) A department has $730 available for monitors. It has already spent $164 on cables and adapters. The cost of each monitor is $84. What is the maximum number of monitors that can be purchased?

c) A computer lab has 240 printer cartridges in storage. Each month, 15 cartridges are used. For how many months will there be more than 90 cartridges remaining?

8. Trace the assignment statements. Write the value stored in each variable after each line is executed.

Statement	Values after execution
x = 5	
y = 3	
z = x * y	
x = z + 2	
y = x - 4	
z = y + x	